

SHORT COMMUNICATION

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AGE-DEPENDENT SERUM SULFATE LEVELS IN CHILDREN AND ADOLESCENTS

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Summary

We measured inorganic sulfate in 100 μ l of deproteinized serum by micro-assay, using the ^{133}Ba precipitation method. Infants and children under 3 yr ($n = 46$), children from 3 to 9 yr of age ($n = 27$) and adolescents (10–18 yr, $n = 12$) in a hospitalized population, and healthy adults ($n = 10$), were studied. An age-dependent curvilinear regression for serum sulfate characterizes the first group: the mean serum sulfate value falls from 0.47 mmol/l on the first day of life to 0.33 mmol/l at 36 months. Thereafter the mean value is about 0.33 mmol/l. The 95% confidence intervals were defined for each group.

Introduction

Most methods for the measurement of serum sulfate require a large sample volume which precludes their use in the pediatric population. Altered serum and extracellular levels of this anion in states of variable nutritional and renal function can only be evaluated if normal values have been established. We have adapted the radio-labelled barium method [1] for microassay of sulfate in 100 μ l of serum [2] and report here, for the first time, the age-dependent serum sulfate concentrations in hospitalized pediatric subjects and healthy ambulant adults.

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Materials and methods

Serum samples

Venous blood samples were taken from in-patients at the Montreal Children's Hospital without concern for time of day or relation to meals. The blood was drawn into tubes without heparin, the serum separated at room temperature, then frozen at -20°C , and analyzed within three to four weeks. Only samples from patients without clinical evidence of renal dysfunction or with normal values for serum urea nitrogen and creatinine were used. Samples were also obtained from healthy adult laboratory personnel, and from eight children of varying age with established renal failure.

Serum sulfate assay

Our modification of the barium precipitation method for microanalysis of serum sulfate has been described elsewhere [2]. In brief, 100 μl of serum is treated with 50 μl of TCA (20% v/v) in methanol (50% v/v) and centrifuged briefly to sediment precipitated protein. To 100 μl of supernatant is added 10 μl of 5 mmol/l Na_2SO_4 followed by 15 μl of 10 mmol/l BaCl_2 containing approximately 2×10^4 dpm ^{133}Ba . The solution is held at 4°C for 3 h then spun in the cold at $9600 \times g \times 1$ h. An aliquot (75 μl) of supernatant is counted for γ -emission. The quantity of sulfate in serum bears a stoichiometric relationship to the decrease in radioactivity of the supernatant.

Statistical

Each value is the average of an assay in triplicate or quadruplicate. We used the Student *t* test to analyze differences between male and female values, and log transformation to derive homogeneous distributions among groups [5]; the means and confidence limits for the various age groups were then calculated. Log transformation allowed us to calculate a curvilinear regression equation and 95% confidence limits in the 0–36 months age group.

Results

We analyzed the sera from 85 children (47 female, 38 male) ranging in age from newborn to 18 years, and from 10 adults (2 female, 8 male). Serum sulfate was not sex dependent in any age group and the values for male and female subjects have been combined for the analysis of age dependence, as shown in Fig. 1 and Table I.

Serum sulfate values are higher in the infant than in later life. Curvilinear regression analysis reveals that the decline in serum sulfate concentration between birth and 3 years of age is significant ($F_{1,44} = 4.93$, $p < 0.05$). The predicted mean value on the first day of life is 0.47 mmol/l with 95% confidence limits between 0.295 mmol/l and 0.945 mmol/l; by 3 years of age the predicted mean value is 0.33 mmol/l for which the 95% confidence limits are 0.225 mmol/l to 0.675 mmol/l. Thereafter, the observed mean value is approximately 0.33 mmol/l in the child, adolescent and adult, in accordance with previous values reported for the adult [1,6].

The concentration of sulfate in eight children with renal failure ranging in

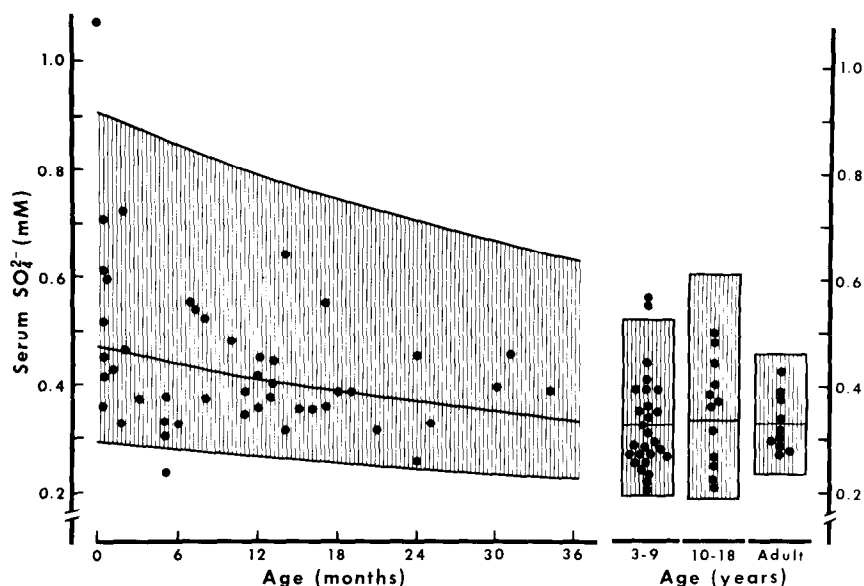


Fig. 1. Age-dependent serum sulfate values in human subjects; means and 95% confidence limits are indicated.

age from newborn to 16 years, was considerably above that in the control population (mean, 0.99 mmol/l; range, 0.68–1.45 mmol/l). All values fell outside the 95% confidence limits for controls without renal disease.

Some variation in serum sulfate can be attributed to dietary protein intake. Sulfate levels in the overnight fasted adult were 0.29 ± 0.03 mmol/l (mean $\pm$ S.D., $n = 10$) whereas random daytime values were significantly higher (0.33 ± 0.06 mmol/l; $t' = 2.07$, $p < 0.05$), more variable ($F_{9,12} = 2.83$, $p < 0.05$) and skewed upwards.

TABLE I
AGE-DEPENDENT SERUM SULFATE CONCENTRATION IN MAN

Age	No. of subjects **	Serum sulfate (mmol/l) ***	
		Mean	95% confidence intervals
Newborn *	46	0.47	0.29–0.95
At 36 mo *		0.33	0.22–0.67
3–9 yr	27	0.32	0.19–0.52
10–18 yr	12	0.33	0.19–0.61
Adult	10	0.33	0.22–0.43

* The fall in serum sulfate concentration between birth and 3 yr observes a curvilinear regression (see Methods and Fig. 1 for explanation). Values stated in table are extrapolations based on the regression equation.

** Data for female and male subjects are pooled, there being no sex dependence in serum sulfate levels.

*** Estimated in samples of serum ($100 \mu\text{l} \times 3$) obtained without control of diet or time of day and analyzed by the ^{133}Ba precipitation method [2].

Discussion

Free sulfate anion in body fluids is derived chiefly from the oxidation of organic sulfur, while a small fraction is contributed by intestinal absorption of sulfate and hydrolysis of sulfoesters. Sulfate is excreted from the body largely in the urine. Serum sulfate thus reflects the equilibration between intake, distribution in tissue pools, and renal excretion. The latter seems to be of particular importance. With progressive renal failure, sulfate levels rise as the rate of glomerular filtration declines [3,4,7,8]. We found serum sulfate concentrations in eight children with varying degrees of renal failure to be above the age-specific 95% confidence limit in keeping with previous observations in adults [7,8]. Thus, in moderate to severe uremia, a considerable fraction of the increased anion gap may be accounted for by sulfate and with the development of long-term management of renal failure in man, interest in sulfate homeostasis has been rekindled.

The elevated serum sulfate values of young infants are of special interest, because glomerular filtration rate has long been known to be low in young infants [9], and because of the analogy to serum phosphate levels which are highest in the young infant [10]. The finding suggests that the renal contribution to sulfate homeostasis parallels that for phosphate anion. However we know from studies in progress that parathyroid hormone does not have a controlling effect on sulfate levels in serum (D.E.C. Cole and C.R. Scriver, unpublished observations).

Availability of free inorganic sulfate influences *in vitro* incorporation of the anion into biologically important sulfate esters, in particular the sulfated mucopolysaccharides [11] and sulfated phenols [12,13]. It may be that elevated serum sulfate levels in the newborn have an important biological function in the development of competent mechanisms for connective tissue synthesis and drug metabolism. The observed inter-individual variation in serum sulfate is greater than that for serum phosphorus. There are three factors which may contribute to this finding: (1) the precision of the microassay for sulfate (CV = 7%, ref. 2) is less than that for phosphorus (CV = 2.7%, Dr. Claire Dupont, personal communication); (2) the normal fluctuations in sulfate influx into the intravascular compartment are likely to be much larger than the analogous fluxes for phosphate; (3) there is no evidence for rapid feedback regulation of elevated sulfate. Further studies of sulfate metabolism will be aided by the method used here to define control values and, in turn, such studies should enhance our understanding of sulfate metabolism and homeostasis in man.

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